1 Lecture 08 LRM Short Answer Solutions

2 Question #1

Let y be a random variable with $\mathrm{E}(y) = \mu$. The expected squared error when a is used as a prediction of y is

$$\mathrm{E}\left[(y-a)^2\right].$$

Show that the value of a that minimizes expected squared prediction error is the expected value of y.

Solution:

Define

$$y = \mu + u,$$

where $\mu = \mathrm{E}(y)$. Then

$$\mathrm{E}(u) = \mathrm{E}(y-\mu) = \mathrm{E}(y) - \mu = 0.$$

Substituting into the expected squared error,

$$\mathrm{E}\left[(y-a)^2\right] = \mathrm{E}\left[(u+\mu-a)^2\right].$$

Expanding the square,

$$\mathrm{E}\left[u^2 + 2u(\mu-a) + (\mu-a)^2\right].$$

Since μ and a are constants,

$$\mathrm{E}\left[(y-a)^2\right] = \mathrm{E}(u^2) + 2(\mu-a)\mathrm{E}(u) + (\mu-a)^2.$$

Because $\mathrm{E}(u) = 0$,

$$\mathrm{E}\left[(y-a)^2\right] = \mathrm{E}(u^2) + (\mu-a)^2.$$

The first term does not depend on a. Therefore, expected squared prediction error is minimized by minimizing the second term, which can be made equal to zero by setting

$$a = \mu.$$

3 Question #2

Let y be a random variable, and let

$$m(x) = \mathrm{E}(y \mid x)$$

denote the conditional expectation function for y.

a. Define

$$y = m(x) + u.$$

Show that

$$\mathrm{E}(u \mid x) = 0.$$

Solution:

Using the definition of u,

$$\mathrm{E}(u \mid x) = \mathrm{E}(y - m(x) \mid x) = \mathrm{E}(y \mid x) - m(x) = 0.$$

b. Let $h(x)$ denote any function of x. Using the result from part (a), show that

$$\mathrm{E}(h(x)u) = 0.$$

Solution:

First, condition on x:

$$\mathrm{E}(h(x)u \mid x) = h(x)\mathrm{E}(u \mid x) = 0.$$

Then, by the law of iterated expectations,

$$\mathrm{E}(h(x)u) = \mathrm{E}_x\left[\mathrm{E}(h(x)u \mid x)\right] = \mathrm{E}_x(0) = 0.$$

c. The expected squared error when the function $g(x)$ is used as a prediction of y is

$$\mathrm{E}\left[(y - g(x))^2\right].$$

Using the result from part (b), show that the function $g(x)$ that minimizes expected squared prediction error is the conditional expectation function $m(x)$.

Solution:

Substituting

$$y = m(x) + u$$

into the expected squared error gives

$$\mathrm{E}\left[(y - g(x))^2\right] = \mathrm{E}\left[(u + m(x) - g(x))^2\right].$$

Expanding the square,

$$\begin{aligned}\mathrm{E}\left[(y - g(x))^2\right] = \mathrm{E}(u^2) &+ 2\mathrm{E}\left[u(m(x) - g(x))\right] \\ &+ \mathrm{E}\left[(m(x) - g(x))^2\right].\end{aligned}$$

Define

$$h(x) = m(x) - g(x).$$

Then the middle term is

$$\mathrm{E}(uh(x)),$$

which equals zero by part (b).

Therefore,

$$\mathrm{E}\left[(y - g(x))^2\right] = \mathrm{E}(u^2) + \mathrm{E}\left[(m(x) - g(x))^2\right].$$

The first term does not depend on $g(x)$. Therefore, expected squared prediction error is minimized by minimizing the second term, which can be made equal to zero by setting

$$g(x) = m(x) = \mathrm{E}(y \mid x).$$

4 Question #3

The OLS estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$.

The OLS solution satisfies the first-order conditions

$$-2\sum_{i=1}^{n} x_{ik}\left(y_i - x_{i1}b_1 - x_{i2}b_2 - \cdots - x_{iK}b_K\right) = 0,$$

for each $k = 1, 2, \ldots, K$.

Define the OLS prediction error as

$$e_i = y_i - x_i'b.$$

a. Suppose $x_{i1} = 1$ for all i, so that the first column of X is a constant. Show that

$$\sum_{i=1}^{n} e_i = 0.$$

Solution:

See the lecture notes.

b. Suppose $x_{i1} = 1$ for all i. Show that

$$\bar{y} = \bar{x}'b,$$

where

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Solution:

See the lecture notes.

c. Show that the OLS prediction errors have zero sample covariance with each explanatory variable:

$$\widehat{\mathrm{Cov}}(x_k, e) = 0, \qquad k = 1, 2, \ldots, K.$$

For this problem, define

$$\widehat{\mathrm{Cov}}(x_k, e) = \frac{1}{n}\sum_{i=1}^{n}(x_{ik} - \bar{x}_k)(e_i - \bar{e}),$$

where

$$\bar{x}_k = \frac{1}{n}\sum_{i=1}^{n} x_{ik}, \qquad \bar{e} = \frac{1}{n}\sum_{i=1}^{n} e_i.$$

Solution:

See the lecture notes.

5 Question #4

The least squares estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$.

Consider the bivariate prediction equation

$$\hat{w} = b_1 + b_2 e.$$

For concreteness, let w denote wage and let e denote education. In this case, X is an $n \times 2$ matrix with a first column of ones and a second column containing the observed values of education:

$$X = \begin{bmatrix} 1 & e_1 \\ 1 & e_2 \\ \vdots & \vdots \\ 1 & e_n \end{bmatrix}, \qquad Y = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}.$$

Define the following sample averages:

$$\bar{e} = \frac{1}{n}\sum_{i=1}^{n} e_i, \qquad \overline{ee} = \frac{1}{n}\sum_{i=1}^{n} e_i^2,$$

$$\bar{w} = \frac{1}{n}\sum_{i=1}^{n} w_i, \qquad \overline{ew} = \frac{1}{n}\sum_{i=1}^{n} e_i w_i.$$

a. Starting from $b = (X'X)^{-1}X'Y$, show that the least squares estimator can be written as

$$b = \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}.$$

In your derivation, explicitly show the matrix multiplication using the elements of e and w that produce $X'X$ and $X'Y$, and then use the definitions of the sample averages above to obtain the final expression.

Solution:

See the lecture notes.

b. Use matrix algebra to show that the slope estimator can be written as

$$b_2 = \frac{\widehat{\mathrm{Cov}}(e, w)}{\widehat{\mathrm{Var}}(e)}.$$

For this problem, define the sample covariance and variance as

$$\widehat{\mathrm{Cov}}(e, w) = \overline{ew} - \bar{e}\bar{w}, \qquad \widehat{\mathrm{Var}}(e) = \overline{ee} - \bar{e}^{\,2}.$$

Solution:

See the lecture notes.